

Java ClassCastException – Lab Report

Experiment	ClassCastException Handling in Java
Language	Java
Exception	ClassCastException

1. What is ClassCastException?

ClassCastException is a runtime exception in Java that occurs when an object is explicitly cast to a class or type that it is not actually an instance of. In this program, an Integer object is stored in an Object reference and then incorrectly cast to String. Java detects the invalid cast at runtime and throws ClassCastException.

2. Objective

To understand ClassCastException in Java and demonstrate how to handle an invalid type casting operation using try-catch.

3. Task Description

Write a Java program that stores an Integer object in an Object reference, attempts to cast it to String, catches the resulting ClassCastException, and displays a meaningful exception message instead of terminating the program abruptly.

4. Step-Wise Execution

1. Start the program by executing the main() method.
2. Create an Object reference named value and store Integer.valueOf(100) in it.
3. Attempt to cast value from Integer to String using (String) value.
4. The object is actually an Integer, so the cast to String is invalid.
5. Java throws a ClassCastException at runtime.
6. The catch block catches the exception.
7. The program prints: "Exception: Invalid type casting".
8. Program execution ends normally after handling the exception.

5. Algorithm

Step 1: Start

Step 2: Enter the try block.

Step 3: Create an Object containing an Integer value.

Step 4: Cast the Object to String.

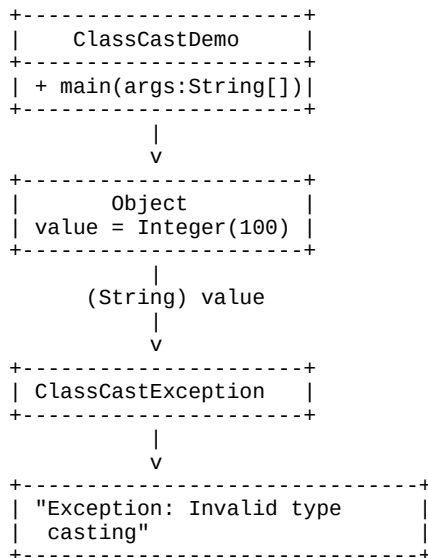
Step 5: If the cast is invalid, ClassCastException occurs.

Step 6: Catch ClassCastException.

Step 7: Display "Exception: Invalid type casting".

Step 8: Stop.

6. UML Diagram



7. Program Code

```
class ClassCastDemo {
    public static void main(String[] args) {
        try {
            Object value = Integer.valueOf(100);
            String text = (String) value;
            System.out.println(text);
        } catch (ClassCastException e) {
            System.out.println("Exception: Invalid type casting");
        }
    }
}
```

8. Code Explanation

Object value = Integer.valueOf(100); — Creates an Integer object containing 100 and stores it in an Object reference.

String text = (String) value; — Attempts to convert/cast the Integer object into a String. This is an invalid cast.

ClassCastException — Java throws this runtime exception because the actual object is Integer, not String.

catch (ClassCastException e) — Catches the exception so the program can handle it safely.

System.out.println(...) — Displays the message "Exception: Invalid type casting".

9. Output Screenshot

Java Program Output

Exception: Invalid type casting

Expected output: Exception: Invalid type casting